# Chapter 2. Basic Properties of Fourier Series

*Uniqueness, convolution, good kernels, Fejer and Poisson*

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## 2.1 Examples and formulation of the problem

**Motivation.** For Riemann integrable $f$ on the circle: in what sense does its Fourier series converge back to $f$?

### 2.1.1 Main definitions and some examples

A bounded $f$ on $[a,b]$ is (Riemann) **integrable** if $\sup_P L(P,f) = \inf_P U(P,f)$; piecewise continuous functions are integrable (Appendix), and complex $f = u + iv$ is integrable if $u, v$ are. A function on the circle is a $2\pi$-periodic function on $\mathbb{R}$, equivalently $F(e^{i\theta}) = f(\theta)$ on $\{|z| = 1\}$; it is integrable on the circle if integrable on one (hence every) period.

**Definition (Fourier coefficients and series)**

For $f$ integrable on $[a,b]$, $L = b - a$, the $n$-th **Fourier coefficient** is

$$\hat{f}(n) = \frac{1}{L}\int_a^b f(x)e^{-2\pi inx/L}\,dx, \qquad n \in \mathbb{Z},$$

and the **Fourier series** is $f(x) \sim \sum_{n=-\infty}^{\infty} \hat{f}(n)e^{2\pi inx/L}$. On the circle ($L = 2\pi$): $\hat{f}(n) = a_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\theta)e^{-in\theta}\,d\theta$. The **$N$-th partial sum** is $S_N(f)(\theta) = \sum_{|n|\le N} \hat{f}(n)e^{in\theta}$. A **trigonometric polynomial** is $\sum_{|n|\le N} c_n e^{in\theta}$, of degree $N$ if $c_{\pm N} \neq 0$.

**Basic question.** In what sense does $S_N(f) \to f$ as $N \to \infty$?

**Example (Sawtooth)**

For $f(\theta) = \frac{\pi - \theta}{2}$ on $0 < \theta < 2\pi$ (extended periodically, $f(0) = 0$): $\hat{f}(0) = 0$ and, integrating by parts, $\hat{f}(n) = \frac{1}{2in}$ for $n \neq 0$. So $f(\theta) \sim \sum_{n\neq 0} \frac{e^{in\theta}}{2in} = \sum_{n\ge 1} \frac{\sin n\theta}{n}$.

**Example**

For $f(\theta) = |\theta|$ on $[-\pi,\pi]$: $\hat{f}(0) = \frac{\pi}{2}$ and $\hat{f}(n) = \frac{-1+(-1)^n}{\pi n^2}$ for $n \neq 0$, so $f(\theta) \sim \frac{\pi}{2} - \frac{4}{\pi}\sum_{n \text{ odd}\ge 1} \frac{\cos n\theta}{n^2}$.

**Example**

For $f(\theta) = \theta$ on $(-\pi,\pi)$: $\hat{f}(0) = 0$, $\hat{f}(n) = \frac{(-1)^{n+1}}{in}$, so $f \sim 2\sum_{n\ge 1}(-1)^{n+1}\frac{\sin n\theta}{n}$.

**Example (Dirichlet kernel)**

$D_N(\theta) = \sum_{|n|\le N} e^{in\theta} = \frac{\sin((N+\frac{1}{2})\theta)}{\sin(\theta/2)}$, so $\hat{D}_N(n) = 1$ for $|n| \le N$, 0 otherwise. *Proof:* with $\omega = e^{i\theta}$, $\sum_0^N \omega^n = \frac{1-\omega^{N+1}}{1-\omega}$, $\sum_{-N}^{-1}\omega^n = \frac{\omega^{-N}-1}{1-\omega}$, so $D_N = \frac{\omega^{-N}-\omega^{N+1}}{1-\omega}$; multiply through by $\omega^{-1/2}$.

**Example (Poisson kernel)**

For $0 \le r < 1$, $P_r(\theta) = \sum_{n\in\mathbb{Z}} r^{|n|}e^{in\theta} = \frac{1-r^2}{1-2r\cos\theta+r^2}$ (sum two geometric series).

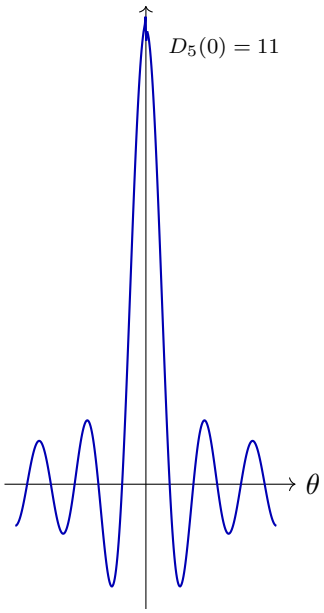



Figure 1: The Dirichlet kernel $D_5$; $\int |D_N|$ grows like $\log N$.

## 2.2 Uniqueness of Fourier series

**Motivation.** Vanishing Fourier coefficients force $f = 0$ at points of continuity, giving uniform convergence when $\sum |\hat{f}(n)| < \infty$.

**Theorem 2.1**

Let $f$ be integrable on the circle with $\hat{f}(n) = 0$ for all $n$. Then $f(\theta_0) = 0$ whenever $f$ is continuous at $\theta_0$.

**Proof.**

*Step 1 (setup).* Let $f$ be real. Translating (which multiplies $\hat{f}(n)$ by a unimodular factor), take $f$ on $[-\pi, \pi]$, $\theta_0 = 0$, and suppose $f(0) > 0$. Then $\int_{-\pi}^{\pi} fp = 0$ for every trigonometric polynomial $p$; we construct $p_k$ with $\int fp_k \to \infty$.

*Step 2 (the polynomials).* Take $0 < \delta \le \pi/2$ with $f > f(0)/2$ on $|\theta| < \delta$; $p(\theta) = \epsilon + \cos\theta$ with $\epsilon > 0$ so small that $|p| < 1 - \epsilon/2$ on $\delta \le |\theta| \le \pi$; $0 < \eta < \delta$ with $p \ge 1 + \epsilon/2$ on $|\theta| < \eta$; $p_k = p^k$; and $|f| \le B$.

*Step 3 (estimates).* On $\delta \le |\theta| \le \pi$: $\left|\int fp_k\right| \le 2\pi B(1 - \epsilon/2)^k \to 0$. On $\eta \le |\theta| < \delta$: $p \ge 0$ and $f > 0$, so $\int fp_k \ge 0$. On $|\theta| < \eta$: $\int fp_k \ge 2\eta \frac{f(0)}{2}(1 + \epsilon/2)^k \to \infty$. So $\int fp_k \to \infty$, a contradiction; thus $f(0) \le 0$, and by $-f$, $f(0) = 0$.

*Step 4 (complex $f$).* $u = \frac{f+\bar{f}}{2}$, $v = \frac{f-\bar{f}}{2i}$ have vanishing coefficients ($\hat{\bar{f}}(n) = \overline{\hat{f}(-n)}$) and are continuous at $\theta_0$. ∎

**Corollary 2.2**

If $f$ is continuous on the circle and $\hat{f}(n) = 0$ for all $n$, then $f = 0$.

**Proof.** Theorem 2.1 at every point. ∎

**Corollary 2.3**

If $f$ is continuous on the circle and $\sum_n |\hat{f}(n)| < \infty$, then $S_N(f) \to f$ uniformly.

**Proof.** By the $M$-test $S_N(f) \to g = \sum \hat{f}(n)e^{in\theta}$ uniformly, $g$ continuous, and termwise integration gives $\hat{g}(n) = \hat{f}(n)$; Corollary 2.2 gives $f = g$. ■

**Corollary 2.4**

If $f$ is twice continuously differentiable on the circle, then $\hat{f}(n) = O(1/|n|^2)$, so the Fourier series of $f$ converges absolutely and uniformly to $f$.

**Proof.** Integrating by parts twice, $2\pi\hat{f}(n) = \frac{1}{in}\int f' e^{-in\theta}d\theta = \frac{-1}{n^2}\int f'' e^{-in\theta}d\theta$, so $|n|^2|\hat{f}(n)| \le \frac{1}{2\pi}\int |f''|$; apply Corollary 2.3. ■

**Remark (on the hypotheses)**

$f \in C^1$ suffices (book exercise), and $f \in C^k$ gives $\hat{f}(n) = o(1/|n|^k)$: smoother $f$, faster decay.

## 2.3 Convolutions

**Motivation.** Partial sums and means are convolutions $f * K$, which multiply Fourier coefficients and smooth.

**Definition (Convolution)**

For $2\pi$-periodic integrable $f, g$, $(f * g)(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)g(x-y)\,dy$. E.g. $S_N(f)(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)\sum_{|n|\le N} e^{in(x-y)}dy = (f * D_N)(x)$.

**Proposition 3.1**

Let $f, g, h$ be $2\pi$-periodic integrable functions. Then:
(i) $f * (g + h) = (f * g) + (f * h)$.
(ii) $(cf) * g = c(f * g) = f * (cg)$ for $c \in \mathbb{C}$.
(iii) $f * g = g * f$.
(iv) $(f * g) * h = f * (g * h)$.
(v) $f * g$ is continuous.
(vi) $\widehat{f * g}(n) = \hat{f}(n)\hat{g}(n)$.

**Lemma 3.2**

Let $f$ be integrable on the circle with $|f| \le B$. There are continuous functions $f_k$ on the circle with $\sup |f_k| \le B$ and $\int_{-\pi}^{\pi} |f(x) - f_k(x)|\,dx \to 0$.

**Proof.** Take $f$ real. Given $\epsilon > 0$ take a partition $-\pi = x_0 < \cdots < x_N = \pi$ with $U(P, f) - L(P, f) < \epsilon$, and $f^* = \sup_{[x_{j-1}, x_j]} f$ on $[x_{j-1}, x_j)$; then $f^* \ge f$ and $\int (f^* - f) \le U - L < \epsilon$. On each $[x_j - \delta, x_j + \delta]$ replace $f^*$ by the segment joining $f^*(x_j - \delta)$ to $f^*(x_j + \delta)$ (periodically at $\pm\pi$). The result $\tilde{f}$ is continuous, $|\tilde{f}| \le B$, and differs from $f^*$ on length $\le 2N\delta$ by at most $2B$; for small $\delta$, $\int |f - \tilde{f}| < 2\epsilon$. Take $\epsilon = 1/k$. ■

**Proof of Proposition 3.1.**

*Step 1 (continuous $f, g$).* (i), (ii): linearity. (iii): substitute $y \mapsto x - y$, use periodicity. (iv): interchange integrals (continuous integrand) and substitute. (v): by uniform continuity, $|f*g(x_1) - f*g(x_2)| \le \sup_y |f(x_1-y) - f(x_2-y)| \sup|g| \to 0$ (using (iii)). (vi):

$$\begin{aligned}\widehat{f*g}(n) &= \frac{1}{2\pi}\int \Big(\frac{1}{2\pi}\int f(y)g(x-y)dy\Big)e^{-inx}dx \\ &= \frac{1}{2\pi}\int f(y)e^{-iny}\Big(\frac{1}{2\pi}\int g(x-y)e^{-in(x-y)}dx\Big)dy = \hat{f}(n)\hat{g}(n).\end{aligned}$$

*Step 2 (approximation).* For integrable $f, g$ take continuous $f_k \to f$, $g_k \to g$ (Lemma 3.2, bound $B$). Then $f*g - f_k*g_k = (f-f_k)*g + f_k*(g-g_k)$, $\sup_x |(f-f_k)*g| \le \frac{B}{2\pi}\int|f-f_k| \to 0$, and likewise the second term. So $f_k * g_k \to f*g$ uniformly, giving (v), and (iii), (iv) in the limit. For (vi), $|\hat{f}(n) - \hat{f}_k(n)| \le \frac{1}{2\pi}\int|f-f_k| \to 0$ and $\widehat{f_k*g_k}(n) \to \widehat{f*g}(n)$. ∎

## 2.4 Good kernels

**Motivation.** Kernels of integral 1, bounded $L^1$ norm and concentrating mass reproduce $f$; $D_N$ fails the middle condition, Fejér and Poisson pass.

**Definition (Good kernels)**

A family $\{K_n\}$ of integrable $2\pi$-periodic functions is a family of **good kernels** if (i) $\frac{1}{2\pi}\int_{-\pi}^{\pi} K_n(x)\,dx = 1$ for all $n$;
(ii) $\int_{-\pi}^{\pi}|K_n(x)|\,dx \le M$ for all $n$;
(iii) for every $\delta > 0$, $\int_{\delta\le|x|\le\pi}|K_n(x)|\,dx \to 0$ as $n \to \infty$. (If $K_n \ge 0$, (ii) follows from (i).)
Likewise for continuous parameters.

**Theorem 4.1**

If $\{K_n\}$ is a family of good kernels and $f$ is integrable on the circle, then $(f*K_n)(x) \to f(x)$ at every point of continuity $x$ of $f$. If $f$ is continuous, the convergence is uniform.

**Proof.** Let $f$ be continuous at $x$, $\epsilon > 0$, and $|f(x-y) - f(x)| < \epsilon$ for $|y| < \delta$. By (i),

$$(f*K_n)(x) - f(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} K_n(y)[f(x-y) - f(x)]\,dy,$$

so with $|f| \le B$,

$$\begin{aligned}|(f*K_n)(x) - f(x)| &\le \frac{1}{2\pi}\int_{|y|<\delta}|K_n||f(x-y)-f(x)| + \frac{1}{2\pi}\int_{\delta\le|y|\le\pi}|K_n||f(x-y)-f(x)| \\ &\le \frac{\epsilon M}{2\pi} + \frac{2B}{2\pi}\int_{\delta\le|y|\le\pi}|K_n(y)|dy,\end{aligned}$$

by (ii), and the last term is $< \epsilon$ for large $n$ by (iii). For continuous $f$, $\delta$ is uniform in $x$. ∎

$\{D_N\}$ satisfies (i) but not (ii): [Supplement] $|D_N(\theta)| \ge \frac{2}{|\theta|}|\sin((N+\frac{1}{2})\theta)|$, and summing over the $2N$ subintervals where $|\sin((N+\frac{1}{2})\theta)| \ge \frac{1}{2}$ gives $\int|D_N| \ge c\log N$. This is the root of the pointwise difficulties.

## 2.5 Cesàro and Abel summability: applications to Fourier series

**Motivation.** Cesàro and Abel means replace $D_N$ by the good Fejér and Poisson kernels: trigonometric polynomials are dense, and the Poisson integral solves Dirichlet's problem.

### 2.5.1 Cesàro means and summation

For $s_n = \sum_{k=0}^{n} c_k$, the **Cesàro mean** is $\sigma_N = \frac{s_0+\cdots+s_{N-1}}{N}$, and $\sum c_k$ is **Cesàro summable** to $\sigma$ if $\sigma_N \to \sigma$. Convergent series are Cesàro summable to their sum; $1-1+1-\ldots$ is Cesàro summable to $\frac{1}{2}$.

### 2.5.2 Fejér's theorem

$\sigma_N(f)(x) = \frac{S_0(f)(x)+\cdots+S_{N-1}(f)(x)}{N} = (f * F_N)(x)$, where $F_N = \frac{D_0+\cdots+D_{N-1}}{N}$ is the **Fejér kernel**.

**Lemma 5.1**

$F_N(x) = \frac{1}{N}\frac{\sin^2(Nx/2)}{\sin^2(x/2)}$, and $\{F_N\}$ is a family of good kernels.

**Proof.** With $\omega = e^{ix}$, $D_n = \frac{\omega^{-n}-\omega^{n+1}}{1-\omega}$, so

$$NF_N = \frac{1}{1-\omega}\Big(\sum_{n=0}^{N-1}\omega^{-n} - \omega\sum_{n=0}^{N-1}\omega^{n}\Big) = \frac{1}{1-\omega}\Big(\frac{1-\omega^{-N}}{1-\omega^{-1}} - \frac{\omega(1-\omega^{N})}{1-\omega}\Big) = \frac{\omega^{-N}-2+\omega^{N}}{(1-\omega)(1-\omega^{-1})},$$

with $\omega^{-N}-2+\omega^{N} = -(2\sin(Nx/2))^2$ and $(1-\omega)(1-\omega^{-1}) = (2\sin(x/2))^2$ (signs match, $NF_N \geq 0$). Good: $F_N \geq 0$; $\frac{1}{2\pi}\int F_N = 1$ (average of $D_n$'s); on $\delta \leq |x| \leq \pi$, $\sin^2(x/2) \geq c_\delta > 0$, so $F_N \leq \frac{1}{Nc_\delta}$. ■

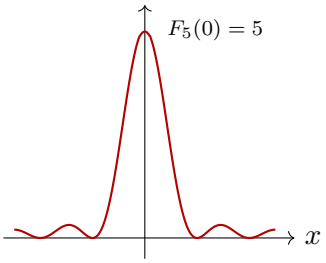



Figure 2: The Fejér kernel $F_5$: non-negative, integral 1, concentrating at 0.

**Theorem 5.2 (Fejér)**

If $f$ is integrable on the circle, its Fourier series is Cesàro summable to $f$ at every point of continuity. If $f$ is continuous, the Fourier series is uniformly Cesàro summable to $f$.

**Proof.** $\sigma_N(f) = f * F_N$ with $\{F_N\}$ good (Lemma 5.1); apply Theorem 4.1. ■

**Corollary 5.3**

If $f$ is integrable on the circle and $\hat{f}(n) = 0$ for all $n$, then $f = 0$ at all points of continuity of $f$.

**Proof.** All partial sums, hence Cesàro means, vanish; apply Theorem 5.2. ■

**Corollary 5.4**

Continuous functions on the circle can be uniformly approximated by trigonometric polynomials.

**Proof.** $\sigma_N(f)$ is a trigonometric polynomial, and $\sigma_N(f) \to f$ uniformly (Theorem 5.2). ■

### 2.5.3 Abel means and summation

$\sum c_k$ is **Abel summable** to $s$ if $A(r) = \sum c_k r^k$ converges for $0 \le r < 1$ and $A(r) \to s$ as $r \to 1^-$. Cesàro summable implies Abel summable to the same limit (not conversely: $\sum (-1)^k (k+1)$ is Abel summable to $\frac{1}{4}$ only). For $f \sim \sum a_n e^{in\theta}$ the **Abel means** are $A_r(f)(\theta) = \sum r^{|n|} a_n e^{in\theta} = (f * P_r)(\theta)$ (interchange, as $|a_n| \le \sup |f|$).

### 2.5.4 The Poisson kernel and Dirichlet's problem in the unit disc

**Lemma 5.5**

If $0 \le r < 1$, then $P_r(\theta) = \frac{1-r^2}{1-2r\cos\theta+r^2}$, and $\{P_r\}_{0\le r<1}$ is a family of good kernels as $r \to 1^-$.

**Proof.** With $\omega = re^{i\theta}$, $P_r = \frac{1}{1-\omega} + \frac{\bar\omega}{1-\bar\omega} = \frac{1-|\omega|^2}{|1-\omega|^2}$. Good: $P_r > 0$; $\frac{1}{2\pi}\int P_r = 1$ termwise; for $\frac{1}{2} \le r < 1$, $\delta \le |\theta| \le \pi$, $1 - 2r\cos\theta + r^2 = (1-r)^2 + 2r(1-\cos\theta) \ge c_\delta > 0$, so $P_r \le \frac{1-r^2}{c_\delta} \to 0$. ■

**Theorem 5.6**

The Fourier series of an integrable function on the circle is Abel summable to $f$ at every point of continuity. If $f$ is continuous, it is uniformly Abel summable to $f$.

**Proof.** $A_r(f) = f * P_r$ with $\{P_r\}$ good; apply Theorem 4.1. ■

**Theorem 5.7**

Let $f$ be integrable on the unit circle and $u(r,\theta) = (f * P_r)(\theta)$ for $0 \le r < 1$. Then:
(i) $u$ is $C^2$ in the unit disc and $\triangle u = 0$.
(ii) If $\theta$ is a point of continuity of $f$, then $u(r,\theta) \to f(\theta)$ as $r \to 1$. If $f$ is continuous, $u(r,\theta) \to f(\theta)$ uniformly as $r \to 1$.
(iii) If $f$ is continuous, then $u$ is the unique solution of $\triangle u = 0$ in the disc that satisfies (ii).

**Proof.**

(i) $u = \sum a_n r^{|n|} e^{in\theta}$, $|a_n| \le B$. For $r \le \rho < 1$ the series and its first and second $r, \theta$ derivatives (factors $|n|^2$) converge uniformly, so $u$ is $C^2$ (indeed $C^\infty$), differentiable termwise, and each $r^{|n|}e^{in\theta}$ is harmonic (§1.2.2).

(ii) Theorem 5.6.

(iii) [Supplement] (Book exercise.) For another solution $v$, $w = u - v$ is harmonic with $w \to 0$ uniformly as $r \to 1$, so extends by 0 continuously to the closed disc. *Maximum principle:* $w_\epsilon = w + \epsilon(x^2+y^2)$ has $\triangle w_\epsilon = 4\epsilon > 0$, so no interior maximum (there $\triangle w_\epsilon \le 0$); hence $w_\epsilon \le \epsilon$. Let $\epsilon \to 0$: $w \le 0$; by $-w$, $w = 0$ (complex $w$: real and imaginary parts). ■

## 2.6 Selected exercises

**Motivation.** Cesàro implies Abel, a Tauberian converse, Abel means at a jump, and why Theorem 5.7(iii) needs uniformity.

**Exercise 2.13 (Convergent ⇒ Abel, Cesàro ⇒ Abel)**

Let $\sum_{k\ge 0} c_k$ have partial sums $s_n = \sum_{k=0}^{n} c_k$ and Cesàro means $\sigma_N = \frac{1}{N}(s_0 + \cdots + s_{N-1})$.
(a) If $s_n \to s$, show that $\sum c_k$ is Abel summable to $s$.
(b) If $\sigma_N \to \sigma$, show that $\sum c_k$ is Abel summable to $\sigma$.

**Solution.**

(a)

*Step 1.* $|s_n| \le M$, so $|c_k| \le 2M$ and $A(r)$ converges absolutely. With $s_{-1} = 0$,

$$\sum_{k=0}^{N} c_k r^k = \sum_{k=0}^{N} (s_k - s_{k-1}) r^k = (1-r) \sum_{k=0}^{N-1} s_k r^k + s_N r^N.$$

As $s_N r^N \to 0$, $A(r) = (1-r)\sum_{k\ge 0} s_k r^k$.

*Step 2.* $(1-r)\sum r^k = 1$ gives $A(r) - s = (1-r)\sum_k (s_k - s) r^k$. With $|s_k - s| < \epsilon$ for $k \ge K$,

$$|A(r) - s| \le (1-r)\, 2MK + \epsilon (1-r) \sum_{k \ge K} r^k \le 2MK(1-r) + \epsilon,$$

so $\limsup_{r\to 1^-} |A(r) - s| \le \epsilon$.

(b)

*Step 1.* Let $t_N = N\sigma_N = s_0 + \cdots + s_{N-1}$ ($t_0 = 0$), $|\sigma_N| \le M$. Then $|t_N| \le MN$, $|s_n| = |t_{n+1} - t_n| \le M(2n+1)$, $|c_n| \le 4M(n+1)$, so $A(r)$ converges absolutely, $s_N r^N \to 0$, and $A(r) = (1-r)\sum_{k\ge 0} s_k r^k$ as in (a).

*Step 2.* Likewise, from $s_k = t_{k+1} - t_k$,

$$\sum_{k=0}^{N} s_k r^k = t_{N+1} r^N + (1-r) \sum_{k=1}^{N} t_k r^{k-1},$$

and $t_{N+1} r^N \to 0$, so

$$A(r) = (1-r)^2 \sum_{k \ge 1} k \sigma_k r^{k-1}.$$

*Step 3.* $(1-r)^2 \sum_{k\ge 1} k r^{k-1} = 1$, so $A(r) - \sigma = (1-r)^2 \sum_k k(\sigma_k - \sigma) r^{k-1}$; with $|\sigma_k - \sigma| < \epsilon$ for $k \ge K$,

$$|A(r) - \sigma| \le (1-r)^2\, 2MK^2 + \epsilon (1-r)^2 \sum_{k \ge K} k r^{k-1} \le 2MK^2 (1-r)^2 + \epsilon,$$

and conclude as in (a). This gives the chain convergent ⇒ Cesàro ⇒ Abel of §2.5.3. ∎

**Exercise 2.14 (A Tauberian theorem)**

If $\sum_{k\ge 0} c_k$ is Cesàro summable to $\sigma$ and $k c_k \to 0$, prove that $\sum c_k$ converges to $\sigma$.

**Solution.**

*Step 1.* With the notation of Exercise 2.13,

$$s_{N-1} - \sigma_N = \frac{1}{N}\sum_{j=0}^{N-1}(s_{N-1} - s_j) = \frac{1}{N}\sum_{j=0}^{N-1}\sum_{k=j+1}^{N-1} c_k = \frac{1}{N}\sum_{k=1}^{N-1} k c_k,$$

as $c_k$ occurs once for each $j < k$.

*Step 2.* $b_k = kc_k \to 0$, $|b_k| \le B$; with $|b_k| < \epsilon$ for $k \ge K$, $\left|\frac{1}{N}\sum_{k=1}^{N-1} b_k\right| \le \frac{KB}{N} + \epsilon < 2\epsilon$ for large $N$.

*Step 3.* So $s_{N-1} = \sigma_N + (s_{N-1} - \sigma_N) \to \sigma$. ∎

**Exercise 2.17 (Abel means at a jump)**

Let $f$ be integrable on the circle with a jump discontinuity at $\theta$: the limits $f(\theta^{\pm}) = \lim_{h\to 0^+} f(\theta \pm h)$ exist. Prove that $A_r(f)(\theta) \to \frac{1}{2}(f(\theta^+) + f(\theta^-))$ as $r \to 1^-$.

**Solution.**

*Step 1.* $P_r$ is even, so $\frac{1}{2\pi}\int_{-\pi}^{0} P_r = \frac{1}{2\pi}\int_0^{\pi} P_r = \frac{1}{2}$ by Lemma 5.5.

*Step 2.* $A_r(f)(\theta) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(y) f(\theta - y)\, dy$, with $\theta - y < \theta$ for $y > 0$. By Step 1,

$$A_r(f)(\theta) - \frac{f(\theta^+) + f(\theta^-)}{2} = \frac{1}{2\pi}\int_0^{\pi} P_r(y)\big[f(\theta - y) - f(\theta^-)\big]dy + \frac{1}{2\pi}\int_{-\pi}^{0} P_r(y)\big[f(\theta - y) - f(\theta^+)\big]dy.$$

*Step 3.* With $|f| \le B$, choose $\delta$ with $|f(\theta - y) - f(\theta^-)| < \epsilon$ for $0 < y < \delta$ and $|f(\theta - y) - f(\theta^+)| < \epsilon$ for $-\delta < y < 0$. As $P_r > 0$,

$$\left|A_r(f)(\theta) - \frac{f(\theta^+) + f(\theta^-)}{2}\right| \le \frac{\epsilon}{2\pi}\int_{-\pi}^{\pi} P_r + \frac{2B}{2\pi}\int_{\delta \le |y| \le \pi} P_r(y)\, dy = \epsilon + o(1) \quad (r \to 1^-),$$

by property (iii) of $\{P_r\}$. ∎

**Exercise 2.18 (Pointwise boundary values do not determine $u$)**

Let $u(r,\theta) = \partial_\theta P_r(\theta)$ for $0 \le r < 1$. Show that $\triangle u = 0$ in the disc and $u(r,\theta) \to 0$ as $r \to 1$ for each $\theta$, but $u \not\equiv 0$.

**Solution.**

*Step 1 ($P$ is harmonic).* With $z = x + iy = re^{i\theta}$,

$$\frac{1+z}{1-z} = \frac{(1+z)(1-\bar z)}{|1-z|^2} = \frac{1 - |z|^2 + (z - \bar z)}{|1-z|^2}, \qquad \text{so} \qquad P_r(\theta) = \operatorname{Re}\frac{1+z}{1-z}.$$

The real part $U$ of a holomorphic $G = U + iV$ is $C^\infty$ and harmonic: $U_x = V_y$, $U_y = -V_x$ give $U_{xx} + U_{yy} = V_{yx} - V_{xy} = 0$.

*Step 2 ($u$ is harmonic).* $\partial_\theta = x\partial_y - y\partial_x$, so $u = xP_y - yP_x$, and $\triangle(xg) = x\triangle g + 2g_x$ gives

$$\triangle u = x\,\triangle P_y + 2P_{xy} - y\,\triangle P_x - 2P_{xy} = x\,\partial_y(\triangle P) - y\,\partial_x(\triangle P) = 0.$$

*Step 3 (boundary limits).* Differentiating the closed form,

$$u(r,\theta) = -\frac{2r(1-r^2)\sin\theta}{(1 - 2r\cos\theta + r^2)^2}.$$

For $\theta \in 2\pi\mathbb{Z}$, $u(r,\theta) = 0$; otherwise the denominator tends to $(2 - 2\cos\theta)^2 > 0$ and the numerator to 0.

*Step 4.* But $u(\frac{1}{2}, \frac{\pi}{2}) = -\frac{3/4}{(5/4)^2} = -\frac{12}{25} \neq 0$. So the convergence is not uniform, else Theorem 5.7(iii) with $f = 0$ would force $u \equiv 0$. ∎